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Toda transformación de Möbius es composición de traslaciones, dilataciones, rotaciones e inversiones

Proposición 1. *Toda transformación de Möbius es una composición de traslaciones, dilataciones, rotaciones e inversiones.*

***Demostración*:** Separamos en dos casos: $c = 0$ y $c \neq 0$:

- Si $c = 0$, entonces $T(z) = {}^{a}\!/_{d} z + {}^{b}\!/_{d} = {}^{a}\!/_{d} \left(z + {}^{b}\!/_{a}\right)$ donde ${}^{a}\!/_{d} = \left|{}^{a}\!/_{d}\right| e^{i \operatorname{Arg}\left({}^{a}\!/_{d}\right)} = re^{i\theta}$ luego $T(z) = re^{i\theta}\left(z + {}^{b}\!/_{a}\right) = \left(S_r \circ R_\theta \circ T_{{}^{b}\!/_{d}}\right)(z)$.
- Si $c \neq 0$, entonces $T(z) = \dfrac{az+b}{cz+d} = \dfrac{(az+b)\,c + ad - ad}{(cz+d)\,c} = \dfrac{a\,(cz+d) + (cb - ad)}{(cz+d)\,c}$

$$\implies T(z) = \frac{a}{c} + \frac{cb - ad}{c\,(cz - d)} = \frac{a}{c} + \underbrace{\frac{cb-ad}{c^2}}_{re^{i\theta}} \cdot \frac{1}{z + {}^{d}\!/_{c}} = \left(T_{{}^{a}\!/_{c}} \circ S_r \circ R_\theta \circ I \circ T_{{}^{d}\!/_{c}}\right)(z).$$

■

Referenciado en

- `transformacion-mobius-circunferencias-generalizadas`

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